Loading the battery dataset

```
%% STEP 1: Download and Load Battery Dataset
dataFile
= matlab.internal.examples.downloadSupportFile("predmaint","batteryagingdata/
singlecell/v1/singleCellLifeTimeData.zip");
unzip(dataFile);
load("singleCellLifeTimeData.mat");

%% STEP 2: Parse and Segment Data
bParser = batteryTestDataParser(data);
bParser.CurrentVariable = 'Current';
bParser.VoltageVariable = 'Voltage';
bParser.TimeVariable = 'DateTime';
bParser.CycleIndexVariable = 'Cycle_Index';
bParser.StepIndexVariable = 'Step_Index';
segmentedRawDataTable = segmentData(bParser);

%% STEP 3: Extract All Valid "Charge" Data for Cycle 1
chargingData = segmentedRawDataTable(...
    segmentedRawDataTable.Cycle_Index == 1 & ...
    segmentedRawDataTable.CyclingPhases == "Charge", :);
chargingData.TimeInSeconds = seconds(chargingData.DateTime -
chargingData.DateTime(1));
```

Compute the time required to reach 80% charge and 100% charge

```
Vmax = 3.6 % when fully charged
```

```
Vmax =
3.6000
```

```
Vmin = 2   % when fully discharged
```

```
Vmin =
2
```

```

% voltage of battery at the 80% and 100% charges
charge_80 = .8*(Vmax - Vmin) + Vmin
```

```
charge_80 =
3.2800
```

```
charge_100 = Vmax
```

```
charge_100 =
3.6000
```

```

% index of when battery is at the 80% and 100% charges
idx_80 = find(chargingData.Voltage >= charge_80, 1)
```

```
idx_80 =
97
```

```
idx_100 = find(chargingData.Voltage >= charge_100, 1)
```

```
idx_100 =
440
```

```
% time when battery is at the 80% and 100% charges
t_80 = chargingData.TimeInSeconds(idx_80)
```

```
t_80 =
24.3477
```

```
t_100 = chargingData.TimeInSeconds(idx_100)
```

```
t_100 =
1.2465e+03
```

```
fprintf('Time in seconds required to reach 80%% charge: %.2f\n', t_80);
```

```
Time in seconds required to reach 80% charge: 24.35
```

```
fprintf('Time in seconds required to reach 100%% charge: %.2f\n', t_100);
```

```
Time in seconds required to reach 100% charge: 1246.49
```